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QUOTATION

62.4 kWp on-grid solar PV system with zero-export protection

Customer	Sample: customer details removed	**Date**	22 October 2026
Site	Factory, northern Thailand	**Validity**	30 days (sample prices)
System	96 panels (6 strings × 16 panels), 50 kW inverter, metal-sheet roof mount	**Currency**	Thai baht (THB)

System overview. 96 LONGi Hi-MO X10 Explorer 650 W monofacial panels (62.4 kWp) connected to a Huawei SUN2000-50K-MC0 inverter in a 1/2/2/1 string layout across MPPT 1–4, the most efficient arrangement for this inverter. Operating voltage is about 603 V. Expected generation is about **89,300 kWh per year**. Includes a zero-export meter and monitoring through the FusionSolar app.

No.	Description	Unit	Qty	Unit price	Amount (THB)
A. Solar panels and inverter					
1	**Solar panels** LONGi Hi-MO X10 Explorer LR7-72HVH-650M 650 W monofacial, 2382 × 1134 × 30 mm	panel	96	3,276.00	314,496.00
2	**Inverter** Huawei SUN2000-50K-MC0 50 kW 3-phase, 4 MPPT / 8 inputs, with AFCI · 5-year warranty	set	1	89,917.30	89,917.30
3	**Zero-export meter** Huawei DTSU666-H 3-phase, CTs included	set	1	2,904.68	2,904.68
4	**Monitoring device** Huawei SDongleB-06 4G + FusionSolar app	set	1	5,004.67	5,004.67
B. Mounting system for metal-sheet roof					
5	**Mounting rail** 4.8 m aluminium rail; 6 rows of 16 panels, 2 rails per row	length	48	516.17	24,776.16
6	**Rail splice** Joins rails into a continuous run along each row	pc	36	35.32	1,271.52
7	**Metal-roof L-feet** L-brackets fixed to the roof purlins at max. 1.2 m spacing, with screws and EPDM seals (incl. 5% spare)	pc	214	34.35	7,350.90
8	**Rail T-bolt M8** Fixes the L-feet to the rails	set	214	9.14	1,955.96
9	**Mid clamp** For 30 mm panel frames, stainless-steel bolts (incl. 5% spare)	pc	189	16.68	3,152.52
10	**End clamp** For 30 mm panel frames, stainless-steel bolts (incl. spare)	pc	26	16.68	433.68
11	**Grounding clip** Bonds each panel frame to the rail	pc	192	5.04	967.68
12	**Roof waterproofing** PU sealant and butyl tape for every fixing point	lot	1	3,412.50	3,412.50
C. DC electrical system					

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No.	Description	Unit	Qty	Unit price	Amount (THB)
13	**PV cable, red** LINK H1Z2Z2-K 1×6 mm² 1.5 kV DC, 100 m roll, EN 50618	roll	3	3,071.51	9,214.53
14	**PV cable, black** LINK H1Z2Z2-K 1×6 mm² 1.5 kV DC, 100 m roll, EN 50618	roll	3	3,071.51	9,214.53
15	**PV connectors** MC4 male/female pairs, 1500 V (incl. spare)	pair	18	28.47	512.46
16	**DC combiner box** Waterproof steel enclosure KJL KBSN 04	set	1	1,251.17	1,251.17
17	**DC breaker** Suntree SL7N-63 4P 32 A 1000 V DC, one per string	pc	6	676.13	4,056.78
18	**DC surge protection (SPD)** BENY BUD40/3 Type II 3P 1000 V DC, one per MPPT	pc	4	657.30	2,629.20
D. AC electrical system					
19	**AC enclosure** Steel enclosure KJL KBSS9004 (44 × 61 × 23 cm)	set	1	936.17	936.17
20	**Inverter output breaker** CHINT NXM-125S MCCB 3P 100 A 25 kA	pc	1	936.39	936.39
21	**AC surge protection (SPD)** Suntree SUP1-40 Type II 4P 40 kA 385 V AC	pc	1	500.47	500.47
22	**AC cable (phases and neutral)** THW 25 mm² (4 cores × 30 m)	m	120	156.14	18,736.80
23	**AC earth cable** THW 16 mm² green/yellow	m	30	99.24	2,977.20
24	**Conduit and fittings** IMC/EMT 1¼" conduit, boxes, clamps and glands	lot	1	16,380.00	16,380.00
E. Earthing system					
25	**Earth rods** 5/8" × 3 m copper rods with clamps and inspection pits	pc	3	649.80	1,949.40
26	**Main earth conductor** 16 mm² green/yellow to the main earth bar	m	50	99.24	4,962.00
27	**Equipotential bonding** 6 mm² green/yellow for structure, rails and enclosures	m	100	38.00	3,800.00
F. Services					
28	**Installation** Mounting, panels, inverter, electrical work and labelling	lot	1	131,040.00	131,040.00
29	**Roof-work safety equipment** Lifeline, roof walkways and fall protection during installation	lot	1	10,500.00	10,500.00
30	**Testing and commissioning** Tests to IEC 62446: polarity, Voc, Isc, insulation resistance, earth resistance and zero-export check	lot	1	15,750.00	15,750.00
31	**PEA documents** PEA grid-connection application with SLD certified by an electrical engineer	lot	1	15,750.00	15,750.00

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No.	Description	Unit	Qty	Unit price	Amount (THB)
32	**Transport** Delivery of equipment to site	lot	1	10,500.00	10,500.00

Equipment subtotal (sections A–E)	**533,700.67**
Services subtotal (section F)	**183,540.00**
Total before VAT	**717,240.67**
VAT 7%	50,206.85
Grand total	**767,447.52**

Equipment prices exclude delivery; transport is shown separately as item 32. Services in section F are subject to 3% withholding tax (5,506.20 THB), which the customer deducts at payment and certifies to the company with a withholding-tax certificate.

Optional items (not included in the total above)

No.	Description	Unit	Qty	Unit price	Amount (THB)
O1	**Soft starter or VFD (optional)** 22 kW 3-phase for the 18 kW compressor motor, with bypass contactor	set	1	34,125.00	34,125.00
O2	**Ice-machine circuit breaker (optional)** MCCB 3P 63 A in the MDB	pc	1	3,822.00	3,822.00

Loads the solar system will supply (site photos)



Photo 1 — Tube-ice machine, the main load of the system

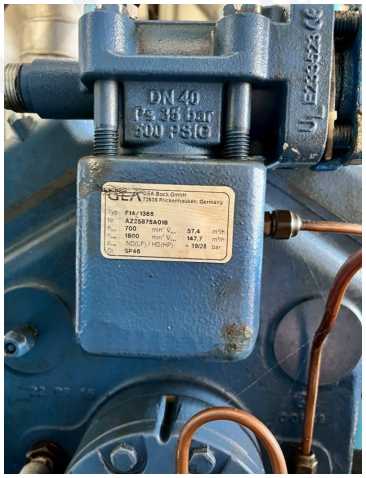


Photo 2 — GEA Bock F14/1366 compressor, swept volume 57.4–147.7 m³/h at 700–1,800 rpm, max. pressure 19/28 bar



Photo 3 — Müller Motor 3-phase induction motor of the auxiliary equipment

The 62.4 kWp system is designed to supply the ice machine and compressor (combined load about 20–22 kW) during the day. Any demand above the solar output is drawn from the PEA grid as normal, and the zero-export meter stops power flowing back into the grid. A soft starter or VFD (item O1) is recommended to reduce inrush current when the compressor starts.

Payment terms

No.	Condition	Share	Amount (THB)
1	On signing the contract	40%	306,979.01
2	When the main equipment (panels, inverter, mounting) is delivered to site	50%	383,723.76
3	After testing and commissioning are complete	10%	76,744.75

General conditions

1. Work period is 45 working days from contract signing and receipt of the first instalment in full (excluding rain days and holidays). The customer prepares the site ready for work to start.
2. Warranty: LONGi panels 15-year product and 30-year performance warranty · Huawei inverter 5 years · installation workmanship 2 years, under the manufacturers' and the company's conditions.
3. Includes the PEA grid-connection application with an SLD certified by an electrical engineer. PEA fees (if any) are not included.
4. Installed on the metal-sheet roof with rails, L-feet and panel clamps; system weight about 13–15 kg/m². The company inspects the purlins and roof structure before installation. If the roof structure needs strengthening or existing roof sheets are damaged, this is quoted separately. Every fixing point is sealed with rubber seals and sealant.
5. Upgrades to the existing MDB are not included. If the site survey shows they are needed, they are quoted separately.
6. Monitoring through the FusionSolar app with automatic alerts. 24-hour monitoring service is not included.
7. Copper cables and certified protection devices are used, and the work is carried out by certified electricians.
8. Equipment prices exclude delivery; delivery to site is shown separately as item 32.

This is a sample quotation published for reference. Customer details have been removed. Prices are indicative for this design as of October 2026; ask Solar ESS for a quotation for your own site.

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